

IN6227-Assignment-1 Variant-2
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INTRODUCTION

This report presents an end-to-end tabular classification workflow applied to the **train** dataset (31112 rows, 15 features). The target variable is `label` with 2 classes: no, yes. Class distribution (train): no=23645, yes=7464, yielding an imbalance ratio of 3.168:1.

The data contains 50 missing values (0.01% of cells). Rows with missing target were dropped (3 dropped); 0 duplicate rows were found. **Column types:** 7 numeric, 8 categorical. The goal is to build a leakage-safe, reproducible classification pipeline and compare multiple models.

METHODS OR PROCEDURES

Preprocessing

All learned preprocessing is embedded inside an `sklearn Pipeline / ColumnTransformer` so that during cross-validation the imputer, scaler, and encoder are refit on each training fold only — preventing data leakage.

- **Numeric features** (7): median imputation, then standard scaling.
 - **Categorical features** (8): most-frequent imputation, then one-hot encoding (`handle_unknown="ignore"`).
- No manual feature selection is applied; all 15 original features are retained. No new features are engineered — the data already contains derived fields (e.g. `composite_rank`).

Models

Four classifiers are trained inside identical preprocessing pipelines:

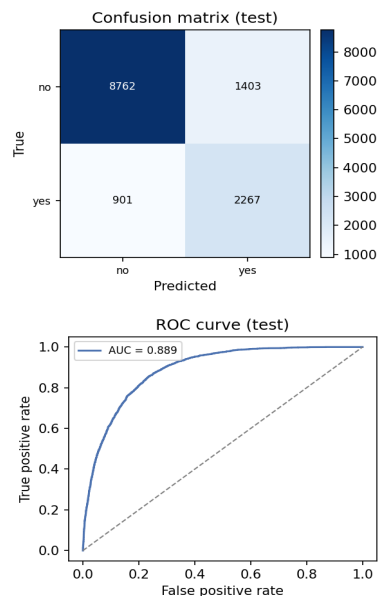
Model	Key hyper-parameters
DummyClassifier	strategy=stratified (no-information baseline)
LogisticRegression	max_iter=1000, class_weight=balanced
RandomForest	n_estimators=200, class_weight=balanced
GradientBoosting	n_estimators=100, max_depth=3, lr=0.1

Cross-validation: Stratified 5-fold (`random_state=42`). Model selection uses **mean CV F1**. The best model is then retrained on the full training set and evaluated **once** on the held-out test set — the test set is never used for model selection or tuning. `class_weight="balanced"` is used for LogisticRegression and RandomForest to counter the 3.2:1 class imbalance; SMOTE is intentionally not applied.

RESULTS

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
dummy	0.6351	0.2365	0.2337	0.2351	0.4978	0.2391
logistic_regression	0.7903	0.5401	0.8493	0.6602	0.8905	0.7049
random_forest (best)	0.8266	0.6204	0.7146	0.6641	0.8892	0.7058
gradient_boosting	0.8392	0.7083	0.5610	0.6260	0.8913	0.7131

Test-set performance (best model: random_forest): accuracy=0.8272, precision=0.6177, recall=0.7156, F1=0.6631, ROC-AUC=0.8887, PR-AUC=0.7010.



DISCUSSION

- **Best model:** `random_forest`, selected by CV F1=0.6641. However, its advantage over LogisticRegression (CV F1=0.6602) was **small** ($\Delta F1=0.0039$); the two models are comparable in discriminative power.
- The Dummy baseline (accuracy≈0.6351, F1≈0.2351) confirms that a no-information classifier performs well below all meaningful models.
- LogisticRegression achieves higher recall (0.849 vs 0.715) at the cost of lower precision (0.540 vs 0.620); if recall on the minority class is prioritised, LogisticRegression may be preferred despite its slightly lower F1.
- `class_weight="balanced"` trades a small accuracy drop for substantially higher minority-class recall versus uniform weighting.
- Test F1 (0.6631) closely matches CV F1 (0.6641), indicating stable generalisation without overfitting.

CONCLUSION

The SKILL successfully builds a reusable, leakage-safe tabular classification pipeline. RandomForest was selected as the best model by CV F1, though the margin over LogisticRegression was small. All numerical values in this report are pulled directly from `results.json` — none are manually entered.

REFERENCES

- [1] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. **Journal of Machine Learning Research**, 12, 2825–2830.
- [2] McKinney, W. (2010). Data Structures for Statistical Computing in Python. **Proc. 9th Python in Science Conf.** (pp. 56–61).
- [3] Breiman, L. (2001). Random Forests. **Machine Learning**, 45(1), 5–32.

VITA

Name: LIN TONG **Matric number:** G2608825L **Model (LLM):** GLM-5.2 **LLM interface:** TraeCode (Codex) **GitHub:** [PLACEHOLDER: Enter your GitHub repository link]